xxxxxxxxx-5253

For the use of a Registered Medical Practitioner or a Hospital or a Laboratory

INTRIL SR 1.5 MG TABLETS

(Indapamide Sustained Release Tablets, 1.5mg)

COMPOSITION

Each sustained release tablet contains Indapamide Ph. Eur. 1.5mg.

DESCRIPTION

White to off-white, round shaped, biconvex film-coated tablet.

Indapamide hemihydrate (4-chloro-N(2-methyl-1-indolinyl)-3-sulphamoyl benzamide hemihydrate) is a non thiazide indole derivative of chlorosulphonamide. The active ingredient is a racemic mixture. Indapamide is a white crystalline lipophilic powder, soluble in methanol, ethanol, acetic acid and ethyl acetate, very slightly soluble in ether, chloroform and benzene and practically insoluble in water. Its molecular weight is 365.84.

CLINICAL PHARMACOLOGY

Indapamide is an oral antihypertensive agent. The mechanism whereby indapamide exerts its antihypertensive action has not been completely elucidated; both vascular and renal actions have been implicated. The possible beneficial pharmacological effects of indapamide in the treatment of hypertension include a reduction in cardiac hyperthrophy and a reduction in the thickening of arterial walls, a prevention of the accumulation of the embryonic isoform of fibronectin in coronary vessels, free radical scavenging leading to stimulation of vasodilator eicosanoid formation, and interaction with renal carbonic anhydrase.

The renal effects of sustained release indapamide hemihydrate are minimal and the antihypertensive effect of indapamide has been attributed to a reduction in vascular reactivity to pressor amines. The finding that indapamide retains its antihypertensive activity to functionally anephric patients lends support to the hypothesis.

The renal site of action of indapamide is the proximal segment of the distal tubule. Indapamide appears to have natriuretic properties (sodium and chloride being excreted in equivalent amounts) with less effect on kaliuresis or uric acid excretion.

Pharmacokinetic properties

Indapamide's kinetics is linear.

Absorption

The fraction of indapamide released is rapidly and totally absorbed via the gastrointestinal digestive tract. Ingestion with food slightly increases the arte and extent of absorption. These changes are unlikely to be clinically significant.

Peak serum level following a single dose occurs about 12 hours after ingestion; repeated administration reduces the variation in serum levels between 2 doses.

Distribution

Indapamide is widely distributed throughout the body, with extensive binding to some specific sites. In blood, it is highly bound to red blood cells (80%) and, more specifically, to carbonic anhydrase (98%) without having any significant inhibiting activity on this enzyme. Binding of indapamide to plasma proteins is 79%. The plasma elimination half life is 14 to 24 hours (mean 18 hours). The drug has a volume of distribution of approximately 60L. Steady state is achieved after 7 days. Repeated administration does not lead to accumulation.

Metabolism & Elimination

The drug is extensively metabolized in the liver, with only 5 to 7% of the dose excreted in the urine as unchanged drug. Elimination is essentially urinary (70% of the dose) and faecal (22%) in the form of inactive metabolites.

High-risk individuals

In patients with severe renal impairment, plasma concentrations sometimes increase significantly.

INDICATIONS

Indapamide is indicated for the treatment of Essential Hypertension

CONTRAINDICATIONS

Severe renal failure, hypokalaemia and hepatic encephalopathy or severe impairment of liver function. Known hypersensitivity to indapamide or to other sulphonamide derivatives.

WARNINGS & PRECAUTIONS:

Special warnings

When liver function is impaired, thiazide-related diuretics may cause hepatic encephalopathy. Administration of the diuretic must be stopped immediately if this occurs.

Special precautions for use

Water and electrolyte balance:

Plasma sodium:

This must be measured before starting treatment, then at regular intervals subsequently. Any diuretic treatment may cause hyponatraemia, sometimes with very serious consequences. The fall in plasma sodium may be asymptomatic initially and regular monitoring in therefore essential, and should be even more frequent in the elderly and cirrhotic patients.

Plasma potassium:

Potassium depletion with hypokalaemia is the major risk of thiazide and related diuretics. The risk of onset of hypokalaemia (<3.4 mmol/l) must be prevented in certain high risk populations, i.e. the elderly, malnourished and/or polymedicated, cirrhotic patients with oedema and ascites ,coronary artery disease and cardiac failure patients. In this situation, hypokalaemia increases the cardiac toxicity of digitalis preparations and the risk of arrhythmias.

Individuals with a long QT interval are also at risk, whether the origin is congenital or iatrogenic. Hypokalaemia, as well as bradycardia, is then a predisposing factor to the onset of severe arrhythmias, in particular, potentially fatal *torsades de pointes*.

More frequent monitoring of plasma potassium is required in all the situations indicated above. The first measurement of plasma potassium should be obtained during the first week following the start of treatment.

Detection of Hypokalaemia requires its correction.

Plasma calcium:

Thiazide and related diuretics may decrease urinary calcium excretion and cause a slight and transitory rise in plasma calcium. Frank hypercalcaemia may be due to previously unrecognized hyperparathyroidism. Treatment should be withdrawn before the investigations of parathyroid function.

Blood glucose:

Monitoring of blood glucose is important in diabetics, in particular in the presence of hypokalaemia.

Uric acid:

Tendency to gout attacks may be increased in hyperuricaemic patients.

Renal function and diuretics:

Thiazide and related diuretics are fully effective only when renal function is normal or only minimally impaired (plasma creatinine below levels of the order of 25 mg/l, i.e. 220 µmol/l in an adult). In the elderly, this plasma creatinine must be adjusted in relation to age, weight and gender.

Hypovolaemia, secondary to the loss of water and sodium induced by the diuretic at the start of treatment causes a reduction in glomerular filtration. This may lead to an increase in blood urea and plasma creatinine. This transitory functional renal insufficiency is of no consequences in individuals with normal renal function but may worsen preexisting renal insufficiency.

Athletes:

The attention of athletes is drawn to the fact that this drug contains an active ingredient which may give a positive reaction in doping test.

Carcinogenicity, Mutagenicity, Impairment of Fertility:

Carcinogenicity studies in mice and rats showed no evidence of tumourigenicity when indapamide was administered in the diet at levels up to 100 mg/kg/day.

Indapamide was negative in mutagenicity tests in bacteria and in a bone marrow mironucleus test in mice.

A reproduction study in rats showed no impairment of male or female fertility at oral indapamide doses up to 25 mg/kg/day; however, the number of implantation sites was reduced at the highest dose. There was a decrease in weight gain of the F1 generation from rats treated orally at 2.5 mg/kg/day. Galactopoiesis was affected in the F1 generation from rats treated orally at 0.5 mg/kg/day and this led to increased mortality of the F2 generation during the first 48 hours of life. No embryo-foetal toxicity nor teratogenic potential were seen in rats (up to 150 mg/kg/day) and in rabbits (up to 180 mg/kg/day).

Use in Pregnancy (Category C)

Thiazides, related diuretics and loop diuretics enter the fetal circulation and may cause electrolyte disturbances. Neonatal thrombocytopenia has been reported with thiazides and related diuretics. Loop diuretics like furosemide and bumetanide are probably also associated with this risk. During the latter part of pregnancy products of this type should only be given on sound indications, and then in the lowest effective dose.

There is no information on the use of indapamide in pregnancy. Whilst animal studies have not suggested any teratogenic effect, indapamide is not recommended for administration to pregnant women unless the expected benefit outweighs the potential risk.

Use in Lactation

It is not known whether indapamide is excreted in breast milk. It is therefore not recommended that the drug be given to lactating women as the possible effect on the newborn is unknown.

Use in Children

Safety and effectiveness have not been established.

ADVERSE REACTIONS

The majority of adverse effects concerning clinical or laboratory parameters are dose- dependent. Thiazides- related diuretics, including indapamide, may cause:

Eye disorders:

Frequency 'not known': Choroidal effusion, acute myopia, acute angle-closure glaucoma

Blood and the lymphatic system disorders:

Very rare: thrombocytopenia, leucopenia, agranulocytosis, aplastic anaemia, haemolytic anaemia.

Nervous system disorders:

Rare: vertigo, fatigue, headache, paresthesia.

Cardiac disorders:

Very rare: arrhythmia, hypotension

Gastrointestinal disorders:

Rare: nausea, constipation, dry mouth.

Very rare: pancreatitis

Hepato-biliary disorders:

In case of hepatic insufficiency, there is a possibility of onset of hepatic encephalopathy (See contra indications and special warnings)

Very rare: abdominal hepatic function

Skin and subcutaneous tissue disorders:

Hypersensitivity reactions, mainly dermatological (common: maculopapular rashes; uncommon: purpura) in subjects with a predisposition to allergic and asthmatic reactions

Possible worsening of pre-existing acute disseminated lupus erythematosus

Laboratory parameters:

During clinical trials, hypokalaemia (plasma potassium<3.4mmol/l) was seen in 10% of patients and <3.2 mmol/l in 4 % of patients after 4 to 6 weeks treatment. After 12 weeks treatment, the mean fall in plasma potassium was 0.23 mmol/l.

Potassium depletion with hypokalaemia particularly serious in certain high risk populations (see special warning and special precautions for use)

Hyponatraemia with hypovolaemia responsible for dehydration and orthostatic hypotension. Concomitant loss of chloride ions may lead to secondary compensatory metabolic alkalosis: the incidence and degree of this effect are slight.

Increase in plasma uric acid and blood glucose during treatment: appropriateness of these diuretics must be very carefully weighed in patients with gout or diabetes.

Very rare: hypercalcaemia.

INTERACTIONS

Interaction with other medicinal products

Combinations that are not recommended

Lithium:

Increased plasma lithium with signs of overdosage, as with a salt-free diet (decreased urinary lithium excretion). However, if the use of diuretics is necessary, careful monitoring of plasma lithium and dose adjustment is required.

Combinations requiring precautions for use

Torsades de pointes-inducing drugs:

Class Ia antiarrhythmics (quinidine, hydroquinidine, disopyramide),

- Class III antiarrhythmics (amiodarone, sotalol, dofetilide, ibutilide),
- Some antipsychotics: phenothiazines (chlorpromazine, cyamemazine,
- levomepromazine, thioridazine, trifluoperazine),

Benzamides (amisulpride, sulpiride, sultopride, tiapride),

Butyrophenones (droperidol, haloperidol);

Others: bepridil, cisapride, diphemanil, erythromycin IV, halofantrine, mizolastine, pentamidine, sparfloxacin, moxifloxacin, vincamine IV.

Increased risk of ventricular arrhythmias, particularly torsades de pointes (hypokalaemia is a risk factor). Monitor for hypokalaemia and correct, if required, before introducing this combination. Clinical, plasma electrolytes and ECG monitoring. Use substances which do not have the disadvantage of causing torsades de pointes in the presence of hypokalaemia.

N.S.A.I.Ds. (systemic route) including COX-2 selective inhibitors, high dose salicylic acid (≥ 3 g/day):

Possible reduction in the antihypertensive effect of indapamide. Risk of acute renal failure in dehydrated patients (decreased glomerular filtration). Hydrate the patient; monitor renal function at the start of treatment.

Angiotensin converting enzyme (A.C.E.) inhibitors:

Risk of sudden hypotension and/or acute renal failure when treatment with an A.C.E. inhibitor is initiated in the presence of preexisting sodium depletion (particularly in patients with renal artery stenosis).

In hypertension, when prior diuretic treatment may have caused sodium depletion, it is necessary:

Either to stop the diuretic 3 days before starting treatment with the A.C.E. inhibitor, and restart a hypokalaemic diuretic if necessary;

Or give low initial doses of the A.C.E. inhibitor and increase the dose gradually.

In congestive heart failure, start with a very low dose of A.C.E. inhibitor, possibly after a reduction in the dose of the concomitant hypokalaemic diuretic.

In all cases, monitor renal function (plasma creatinine) during the first weeks of treatment with an A.C.E. inhibitor.

Other compounds causing hypokalaemia: amphotericin B (IV) gluco-and mineralo-corticoids (systemic route), tetracosactide, stimulant laxatives:

Increased risk of hypokalaemia (additive effect). Monitoring of plasma potassium and correction if required. Must be particularly borne in mind in case of concomitant digitalis treatment. Use non-stimulant laxatives.

Baclofen:

Increased antihypertensive effect. Hydrate the patient; monitor renal function at the start of treatment.

Digitalis preparations:

Hypokalaemia predisposing to the toxic effects of digitalis. Monitoring of plasma potassium and ECG and, if necessary, adjust the treatment.

Combinations to be taken into consideration:

Potassium-sparing diuretics (amiloride, spironolactone, triamterene):

Whilst rational combinations are useful in some patients, hypokalaemia or hyperkalaemia (particularly in patients with renal failure or diabetes) may still occur. Plasma potassium and ECG should be monitored and, if necessary, treatment reviewed.

Metformin:

Increased risk of metformin induced lactic acidosis due to the possibility of functional renal failure associated with diuretics and more particularly with loop diuretics. Do not use metformin when plasma creatinine exceeds 15 mg/l (135 µmol/1) in men and 12 mg/l (110 µmol/1) in women.

Iodinated contrast media:

In the presence of dehydration caused by diuretics, increased risk of acute renal failure, in particular when large doses of iodinated contrast media are used. Rehydration before administration of the iodinated compound.

Imipramine-like antidepressants, neuroleptics:

Antihypertensive effect and increased risk of orthostatic hypotension increased (additive effect).

Calcium (salts):

Risk of hypercalcaemia resulting from decreased urinary elimination of calcium.

Cyclosporin, tacrolimus:

Risk of increased plasma creatinine without any change in circulating cyclosporin levels, even in the absence of water/sodium depletion.

Corticosteroids, tetracosactide (systemic route):

Decreased antihypertensive effect (water/sodium retention due to corticosteroids).

OVERDOSAGE & TREATMENT

Indapamide has been found free of toxicity at up to 40 mg, i.e. 27 times the therapeutic dose. Signs of acute poisoning take the form above all the water/electrolyte disturbances (hyponatraemia, hypokalaemia). Clinically, possibility of nausea, vomiting, hypotension, cramps, vertigo, drowsiness, confusion, polyuria or oliguria possibly to the point of anuria (by hypovolaemia). Initial measures involve the rapid elimination of the ingested substance(s) by gastric wash out and /or administration of activated charcoal, followed by restoration of water/electrolyte balance to normal in a specialized centre.

DOSAGE AND ADMINISTRATION

Oral administration

One tablet per 24 hours, preferably in the morning, to be swallowed whole swallowed with water and not chewed.

At higher dose the antihypertensive action of indapamide is not enhanced but the saluretic effect is increased.

PRESENTATION

Intril SR 1.5 mg tablets are packed in ALU-PVC blister of 10 tablets. Such blisters containing 10 tablets are packed into a carton of 10's, 30's and 100's. Not all presentations may be available locally.

STORAGE

Store in a dry place, below 30°C.

DATE OF REVISION:

15.12.2022

torrent PHARMA

Manufactured by :
TORRENT PHARMACEUTICALS LTD.
Indrad-382 721, Dist. Mehsana, INDIA.

PRODUCT NAME :	Intril SR 1.5 mg tablets	**COUNTRY :** Malysia	**LOCATION :** Chhatral	**Supersedes A/W No.:**			
ITEM / PACK :	Insert	**NO. OF COLORS:** 1	**REMARK : Folded Size 30 mm**				
DESIGN STYLE :	Front-Back	**PANTONE SHADE NOS.:**	**SUBSTRATE :**				
CODE :	xxxxxxxxx-5253	Black	**Activities**	**Department**	**Name**	**Signature**	**Date**
DIMENSIONS (MM) :	180 x 240		Prepared By	**Pkg.Dev**			
ART WORK SIZE :	S/S		Reviewed By	**Pkg.Dev**			
DATE :	15-12-2022		Approved By	**Quality**			